

EXCEL ASSIGNMENT 2

DATA CLEANING AND TRANSFORMATION



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Course: Data Analytics



Assignment: Excel Assignment 2 – Data Cleaning and Transformation



Date: 30 July 2026

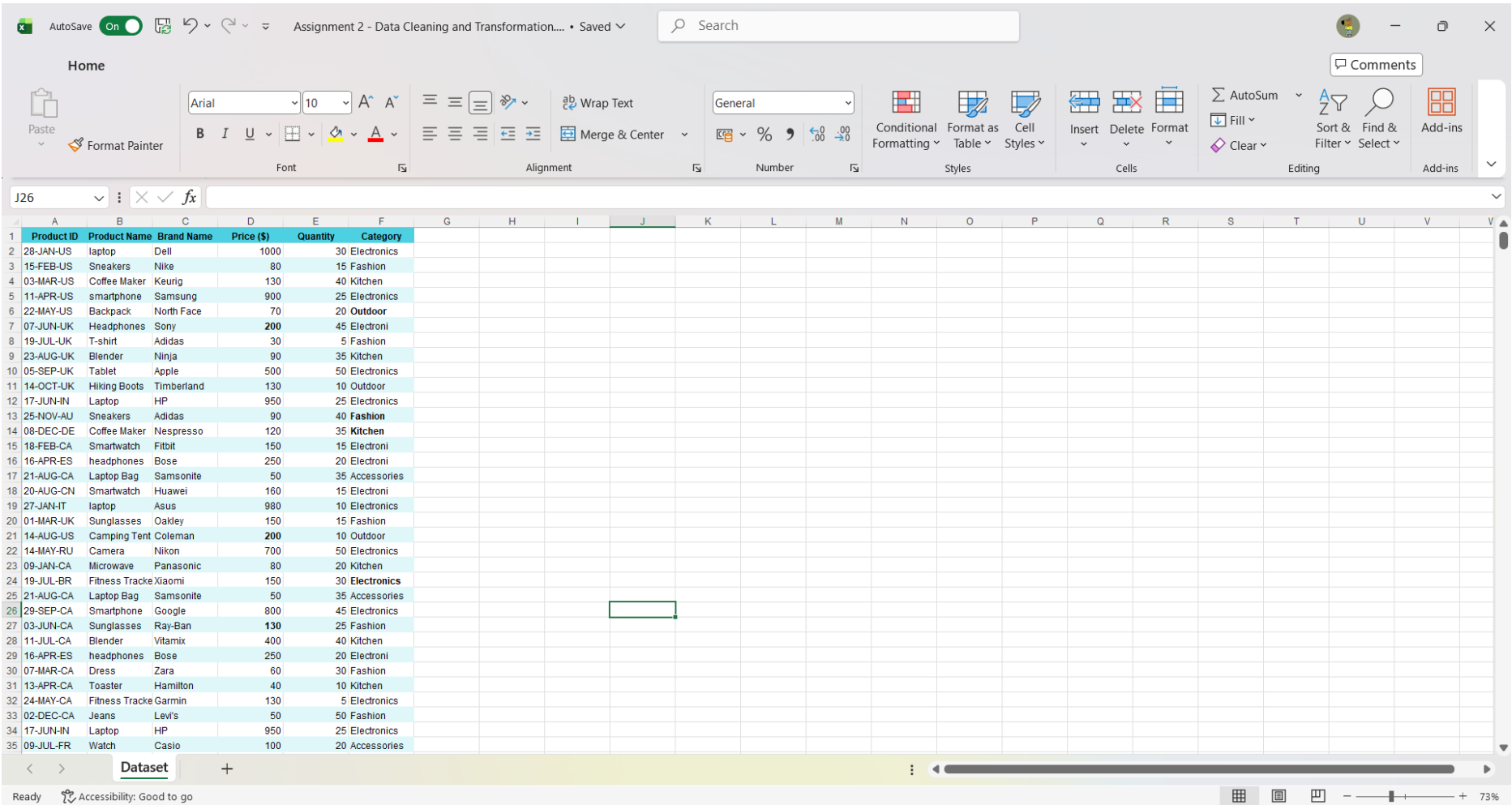
Figure 1: Original Dataset

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V
1	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category																
2	28-JAN-US	laptop	Dell	1000	30	Electronics																
3	15-FEB-US	Sneakers	Nike	80	15	Fashion																
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen																
5	11-APR-US	smartphone	Samsung	900	25	Electronics																
6	22-MAY-US	Backpack	North Face	70	20																	
7	07-JUN-UK	Headphones	Sony		45	Electroni																
8	19-JUL-UK	T-shirt	Adidas	30	5	Fashion																
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen																
10	05-SEP-UK	Tablet	Apple	500	50	Electronics																
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor																
12	17-JUN-IN	Laptop	HP	950	25	Electronics																
13	25-NOV-AU	Sneakers	Adidas	90	40																	
14	08-DEC-DE	Coffee Maker	Nespresso	120	35																	
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electroni																
16	16-APR-ES	headphones	Bose	250	20	Electroni																
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories																
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electroni																
19	27-JAN-IT	laptop	Asus	980	10	Electronics																
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion																
21	14-AUG-US	Camping Tent	Coleman		10	Outdoor																
22	14-MAY-RU	Camera	Nikon	700	50	Electronics																
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen																
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30																	
25	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories																
26	29-SEP-CA	Smartphone	Google	800	45	Electronics																
27	03-JUN-CA	Sunglasses	Ray-Ban		25	Fashion																
28	11-JUL-CA	Blender	Vitamix	400	40	Kitchen																
29	16-APR-ES	headphones	Bose	250	20	Electroni																
30	07-MAR-CA	Dress	Zara	60	30	Fashion																
31	13-APR-CA	Toaster	Hamilton	40	10	Kitchen																
32	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics																
33	02-DEC-CA	Jeans	Levi's	50	50	Fashion																
34	17-JUN-IN	Laptop	HP	950	25	Electronics																
35	09-JUL-FR	Watch	Casio	100	20	Accessories																

Observation:

The original dataset contains **Product ID, Product Name, Brand Name, Price, Quantity, and Category**. It serves as the raw data before any cleaning or transformation. The dataset contains missing values, inconsistent text formatting, and duplicate records that need to be cleaned before analysis.

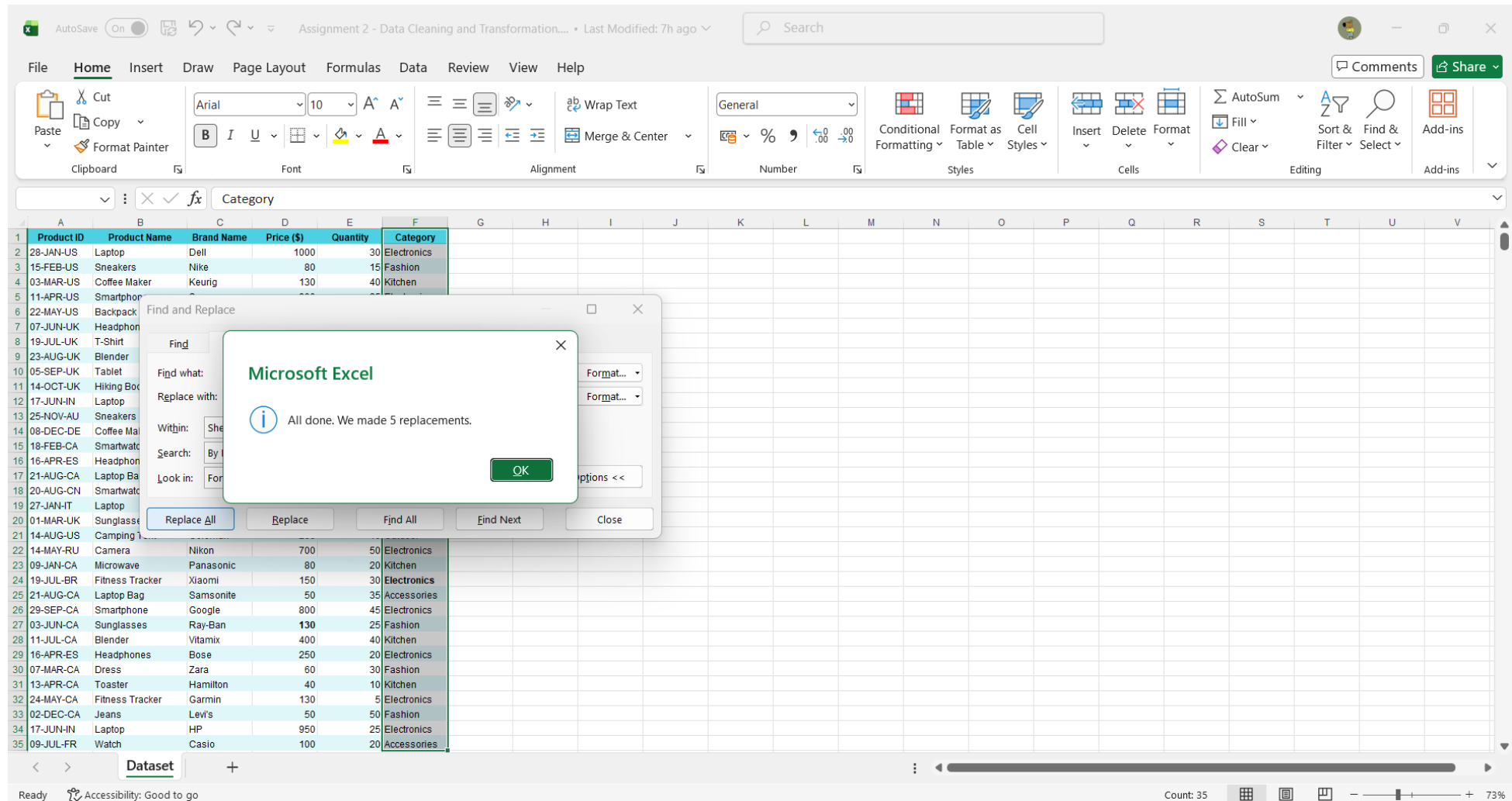
Figure 2: Missing Value Handling



Observation:

The **Price** column was checked for missing values, and the missing price entries were filled with appropriate values to ensure data completeness. This step helps improve data quality and prevents errors during further analysis.

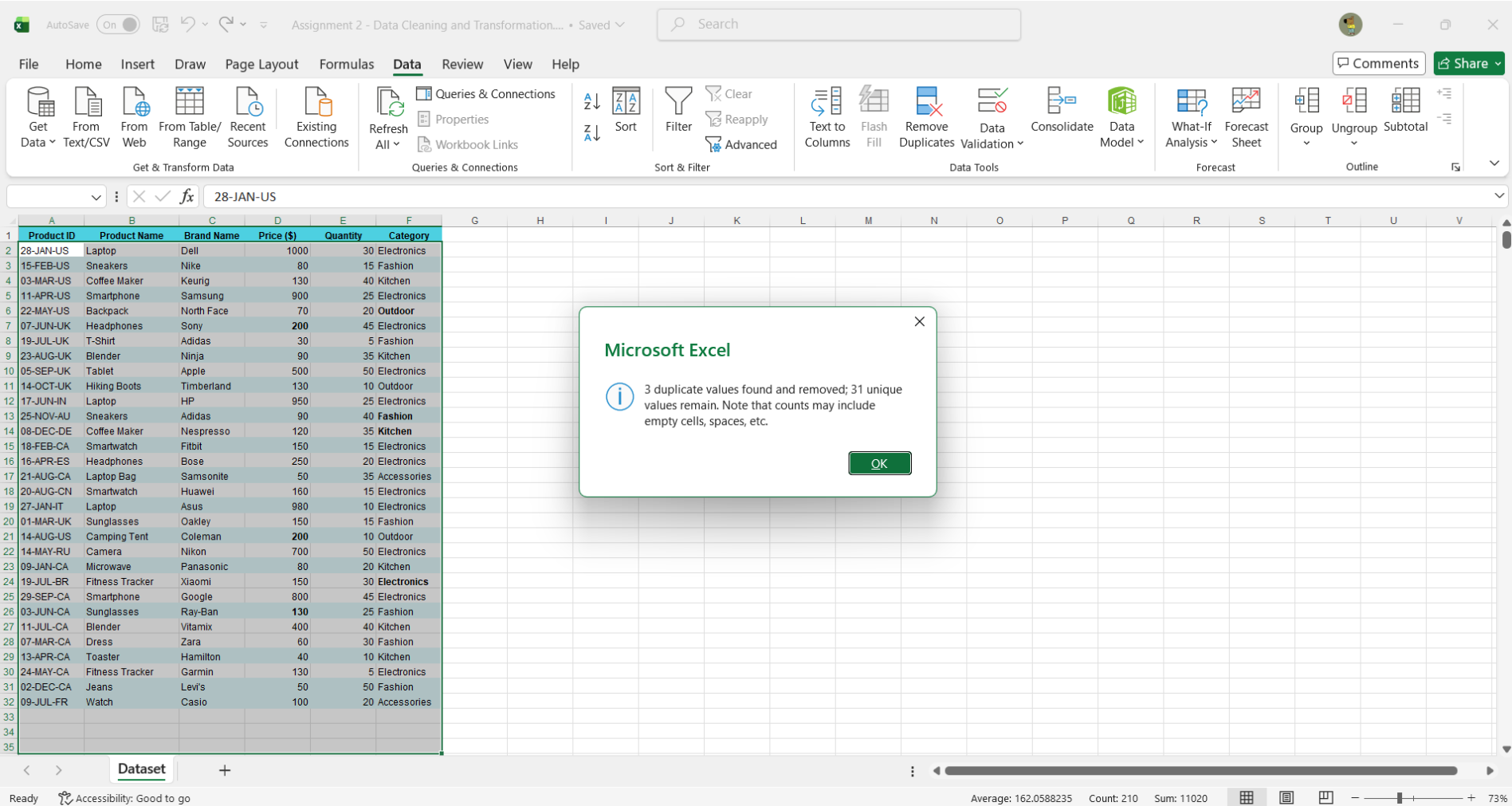
Figure 3: Find and Replace Corrections



Observation:

The **Find and Replace** feature was used to correct the misspelled category value "**Electroni**" to "**Electronics**". Excel successfully made **5 replacements**, ensuring consistent category names throughout the dataset.

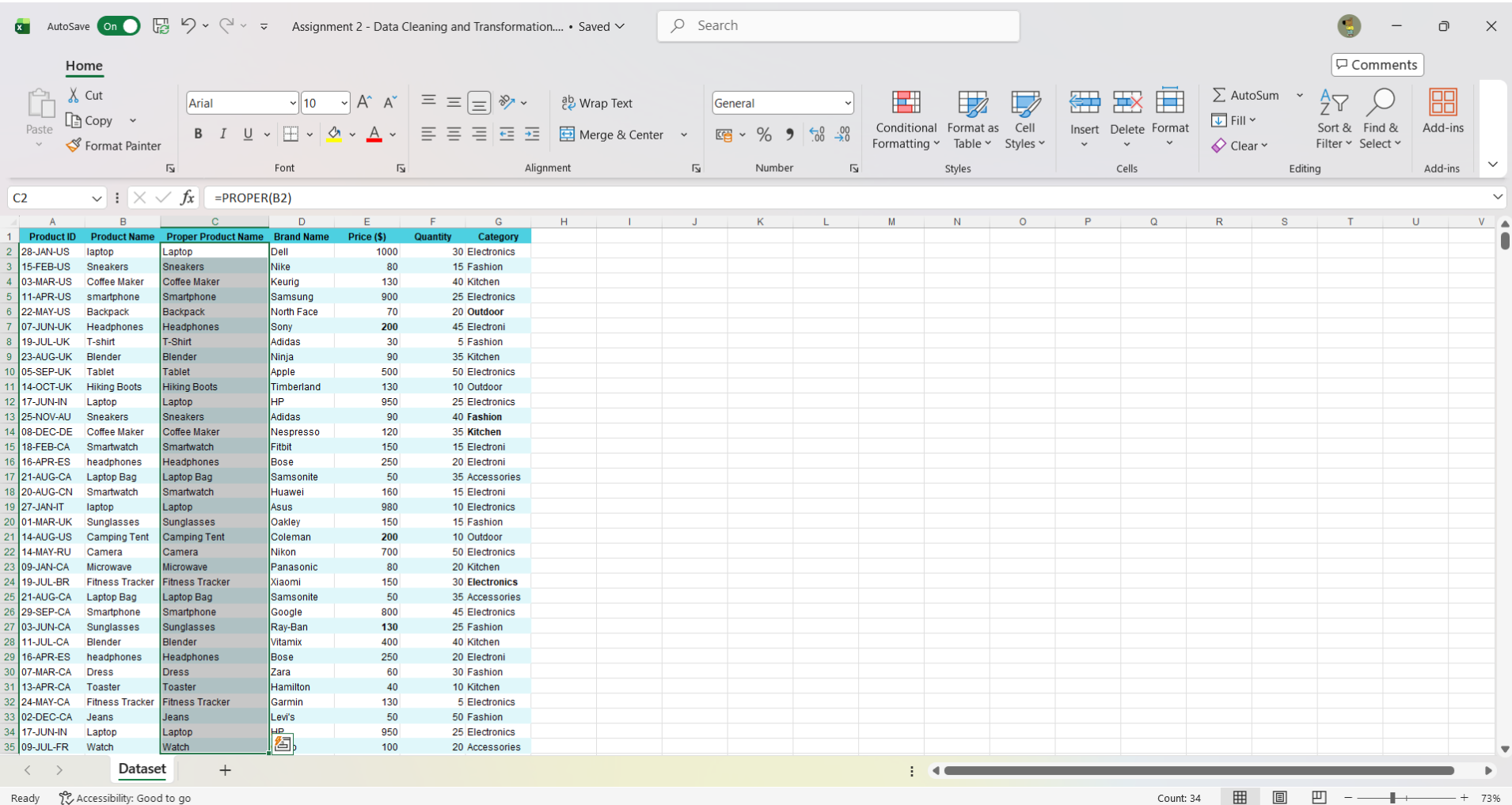
Figure 4: Duplicate Removal



Observation:

The **Remove Duplicates** feature was used to identify and eliminate duplicate records from the dataset. Excel removed **3 duplicate rows**, leaving **31 unique records**. This improved the dataset by ensuring that only unique entries remained for further analysis.

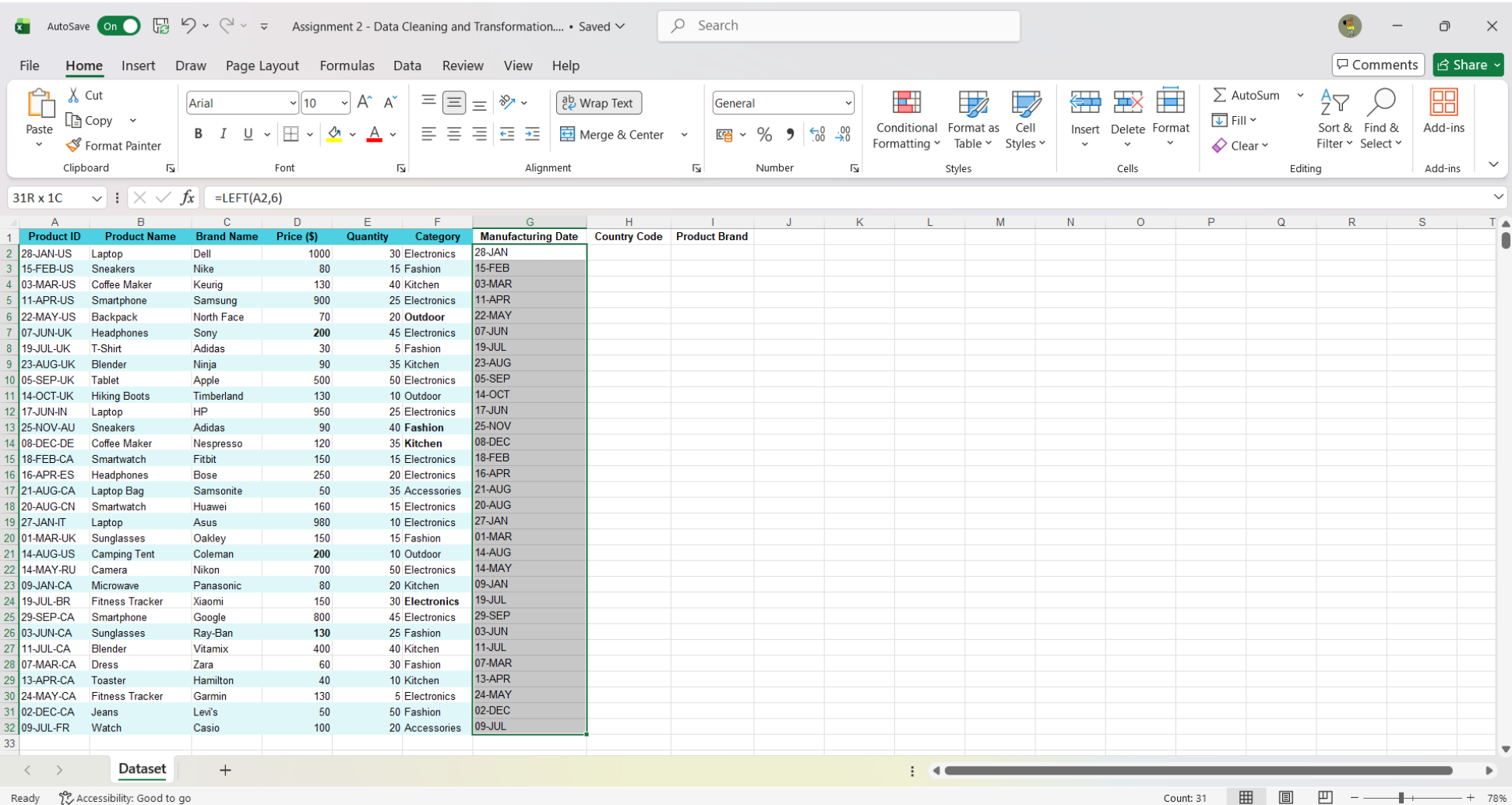
Figure 5: Product Name Standardization (PROPER Function)



Observation:

The **PROPER()** function was used to standardize the text format of the Product Name column. It converted product names to proper case by capitalizing the first letter of each word and converting the remaining letters to lowercase. This ensured a consistent naming format throughout the dataset.

Figure 6: Manufacturing Date Extraction (LEFT Function)



Observation:

The **LEFT()** function was used to extract the first six characters from the **Product ID** (e.g., 28-JAN-US → 28-JAN). This extracted the **day and month portion**, which served as the initial step in creating the **Manufacturing Date** column before converting it into the required **DD-MM-YYYY** format.

Figure 7: Country Code Extraction (RIGHT Function)

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H2 =RIGHT(A2,2)

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Manufacturing Date	Country Code	Product Brand										
1	28-JAN-US	Laptop	Dell	\$1,000.00	30	Electronics	28-01-2026	US	Dell Laptop										
2	15-FEB-US	Sneakers	Nike	\$80.00	15	Fashion	15-02-2026	US	Nike Sneakers										
3	03-MAR-US	Coffee Maker	Keurig	\$130.00	40	Kitchen	03-03-2026	US	Keurig Coffee Maker										
4	11-APR-US	Smartphone	Samsung	\$900.00	25	Electronics	11-04-2026	US	Samsung Smartphone										
5	22-MAY-US	Backpack	North Face	\$70.00	20	Outdoor	22-05-2026	US	North Face Backpack										
6	07-JUN-UK	Headphones	Sony	\$200.00	45	Electronics	07-06-2026	UK	Sony Headphones										
7	19-JUL-UK	T-Shirt	Adidas	\$30.00	5	Fashion	19-07-2026	UK	Adidas T-Shirt										
8	23-AUG-UK	Blender	Ninja	\$90.00	35	Kitchen	23-08-2026	UK	Ninja Blender										
9	05-SEP-UK	Tablet	Apple	\$500.00	50	Electronics	05-09-2026	UK	Apple Tablet										
10	14-OCT-UK	Hiking Boots	Timberland	\$130.00	10	Outdoor	14-10-2026	UK	Timberland Hiking Boots										
11	17-JUN-IN	Laptop	HP	\$950.00	25	Electronics	17-06-2026	IN	HP Laptop										
12	25-NOV-AU	Sneakers	Adidas	\$90.00	40	Fashion	25-11-2026	AU	Adidas Sneakers										
13	08-DEC-DE	Coffee Maker	Nespresso	\$120.00	35	Kitchen	08-12-2026	DE	Nespresso Coffee Maker										
14	18-FEB-CA	Smartwatch	Fitbit	\$150.00	15	Electronics	18-02-2026	CA	Fitbit Smartwatch										
15	16-APR-ES	Headphones	Bose	\$250.00	20	Electronics	16-04-2026	ES	Bose Headphones										
16	21-AUG-CA	Laptop Bag	Samsonite	\$50.00	35	Accessories	21-08-2026	CA	Samsonite Laptop Bag										
17	20-AUG-CN	Smartwatch	Huawei	\$160.00	15	Electronics	20-08-2026	CN	Huawei Smartwatch										
18	27-JAN-IT	Laptop	Asus	\$980.00	10	Electronics	27-01-2026	IT	Asus Laptop										
19	01-MAR-UK	Sunglasses	Oakley	\$150.00	15	Fashion	01-03-2026	UK	Oakley Sunglasses										
20	14-AUG-US	Camping Tent	Coleman	\$200.00	10	Outdoor	14-08-2026	US	Coleman Camping Tent										
21	14-MAY-RU	Camera	Nikon	\$700.00	50	Electronics	14-05-2026	RU	Nikon Camera										
22	09-JAN-CA	Microwave	Panasonic	\$80.00	20	Kitchen	09-01-2026	CA	Panasonic Microwave										
23	19-JUL-BR	Fitness Tracker	Xiaomi	\$150.00	30	Electronics	19-07-2026	BR	Xiaomi Fitness Tracker										
24	25-SEP-CA	Smartphone	Google	\$800.00	45	Electronics	25-09-2026	CA	Google Smartphone										
25	03-JUN-CA	Sunglasses	Ray-Ban	\$130.00	25	Fashion	03-06-2026	CA	Ray-Ban Sunglasses										
26	11-JUL-CA	Blender	Vitamix	\$400.00	40	Kitchen	11-07-2026	CA	Vitamix Blender										
27	07-MAR-CA	Dress	Zara	\$60.00	30	Fashion	07-03-2026	CA	Zara Dress										
28	13-APR-CA	Toaster	Hamilton	\$40.00	10	Kitchen	13-04-2026	CA	Hamilton Toaster										
29	24-MAY-CA	Fitness Tracker	Garmin	\$130.00	5	Electronics	24-05-2026	CA	Garmin Fitness Tracker										
30	02-DEC-CA	Jeans	Levi's	\$50.00	50	Fashion	02-12-2026	CA	Levi's Jeans										
31	09-JUL-FR	Watch	Casio	\$100.00	20	Accessories	09-07-2026	FR	Casio Watch										
32																			
33																			

Ready Accessibility: Good to go Count: 31 78%

Observation:

The **RIGHT()** function was used to extract the last two characters from the **Product ID** column to create the **Country Code** column. This separated the country code (such as **US**, **UK**, **IN**, **CA**, etc.) from the Product ID, making the dataset more structured and easier to analyze.

Figure 8: Product Brand Creation (Concatenation)

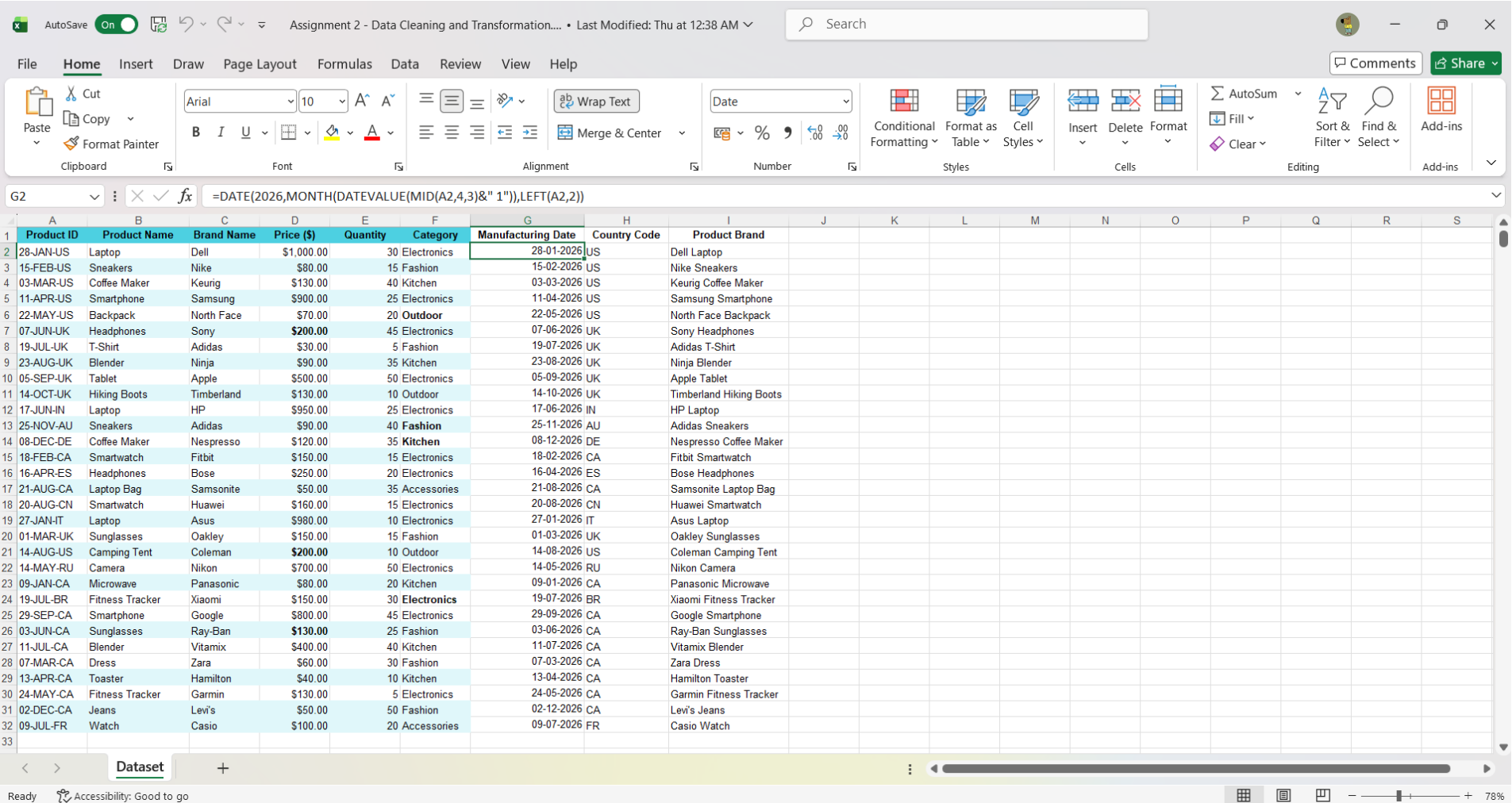
The screenshot shows an Excel spreadsheet titled "Assignment 2 - Data Cleaning and Transformation...". The formula bar displays the formula `=C2 & " " & B2` for cell I2. The spreadsheet contains a dataset with the following columns: Product ID, Product Name, Brand Name, Price (\$), Quantity, Category, Manufacturing Date, Country Code, and Product Brand. The Product Brand column is populated with the concatenated values of Brand Name and Product Name, separated by a space.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
	Product ID	Product Name	Brand Name	Price (\$)	Quantity	Category	Manufacturing Date	Country Code	Product Brand										
2	28-JAN-US	Laptop	Dell	1000	30	Electronics	28-JAN	US	Dell Laptop										
3	15-FEB-US	Sneakers	Nike	80	15	Fashion	15-FEB	US	Nike Sneakers										
4	03-MAR-US	Coffee Maker	Keurig	130	40	Kitchen	03-MAR	US	Keurig Coffee Maker										
5	11-APR-US	Smartphone	Samsung	900	25	Electronics	11-APR	US	Samsung Smartphone										
6	22-MAY-US	Backpack	North Face	70	20	Outdoor	22-MAY	US	North Face Backpack										
7	07-JUN-UK	Headphones	Sony	200	45	Electronics	07-JUN	UK	Sony Headphones										
8	19-JUL-UK	T-Shirt	Adidas	30	5	Fashion	19-JUL	UK	Adidas T-Shirt										
9	23-AUG-UK	Blender	Ninja	90	35	Kitchen	23-AUG	UK	Ninja Blender										
10	05-SEP-UK	Tablet	Apple	500	50	Electronics	05-SEP	UK	Apple Tablet										
11	14-OCT-UK	Hiking Boots	Timberland	130	10	Outdoor	14-OCT	UK	Timberland Hiking Boots										
12	17-JUN-IN	Laptop	HP	950	25	Electronics	17-JUN	IN	HP Laptop										
13	25-NOV-AU	Sneakers	Adidas	90	40	Fashion	25-NOV	AU	Adidas Sneakers										
14	08-DEC-DE	Coffee Maker	Nespresso	120	35	Kitchen	08-DEC	DE	Nespresso Coffee Maker										
15	18-FEB-CA	Smartwatch	Fitbit	150	15	Electronics	18-FEB	CA	Fitbit Smartwatch										
16	16-APR-ES	Headphones	Bose	250	20	Electronics	16-APR	ES	Bose Headphones										
17	21-AUG-CA	Laptop Bag	Samsonite	50	35	Accessories	21-AUG	CA	Samsonite Laptop Bag										
18	20-AUG-CN	Smartwatch	Huawei	160	15	Electronics	20-AUG	CN	Huawei Smartwatch										
19	27-JAN-IT	Laptop	Asus	980	10	Electronics	27-JAN	IT	Asus Laptop										
20	01-MAR-UK	Sunglasses	Oakley	150	15	Fashion	01-MAR	UK	Oakley Sunglasses										
21	14-AUG-US	Camping Tent	Coleman	200	10	Outdoor	14-AUG	US	Coleman Camping Tent										
22	14-MAY-RU	Camera	Nikon	700	50	Electronics	14-MAY	RU	Nikon Camera										
23	09-JAN-CA	Microwave	Panasonic	80	20	Kitchen	09-JAN	CA	Panasonic Microwave										
24	19-JUL-BR	Fitness Tracker	Xiaomi	150	30	Electronics	19-JUL	BR	Xiaomi Fitness Tracker										
25	29-SEP-CA	Smartphone	Google	800	45	Electronics	29-SEP	CA	Google Smartphone										
26	03-JUN-CA	Sunglasses	Ray-Ban	130	25	Fashion	03-JUN	CA	Ray-Ban Sunglasses										
27	11-JUL-CA	Blender	Vitamix	400	40	Kitchen	11-JUL	CA	Vitamix Blender										
28	07-MAR-CA	Dress	Zara	60	30	Fashion	07-MAR	CA	Zara Dress										
29	13-APR-CA	Toaster	Hamilton	40	10	Kitchen	13-APR	CA	Hamilton Toaster										
30	24-MAY-CA	Fitness Tracker	Garmin	130	5	Electronics	24-MAY	CA	Garmin Fitness Tracker										
31	02-DEC-CA	Jeans	Levi's	50	50	Fashion	02-DEC	CA	Levi's Jeans										
32	09-JUL-FR	Watch	Casio	100	20	Accessories	09-JUL	FR	Casio Watch										

Observation:

The **Brand Name** and **Product Name** columns were merged into a new **Product Brand** column using the concatenation formula: `=C2 & " " & B2`. This combined both values into a single field (for example, **Dell Laptop**, **Nike Sneakers**, and **Samsung Smartphone**), creating a more descriptive product identifier and satisfying the assignment requirement to merge the two columns.

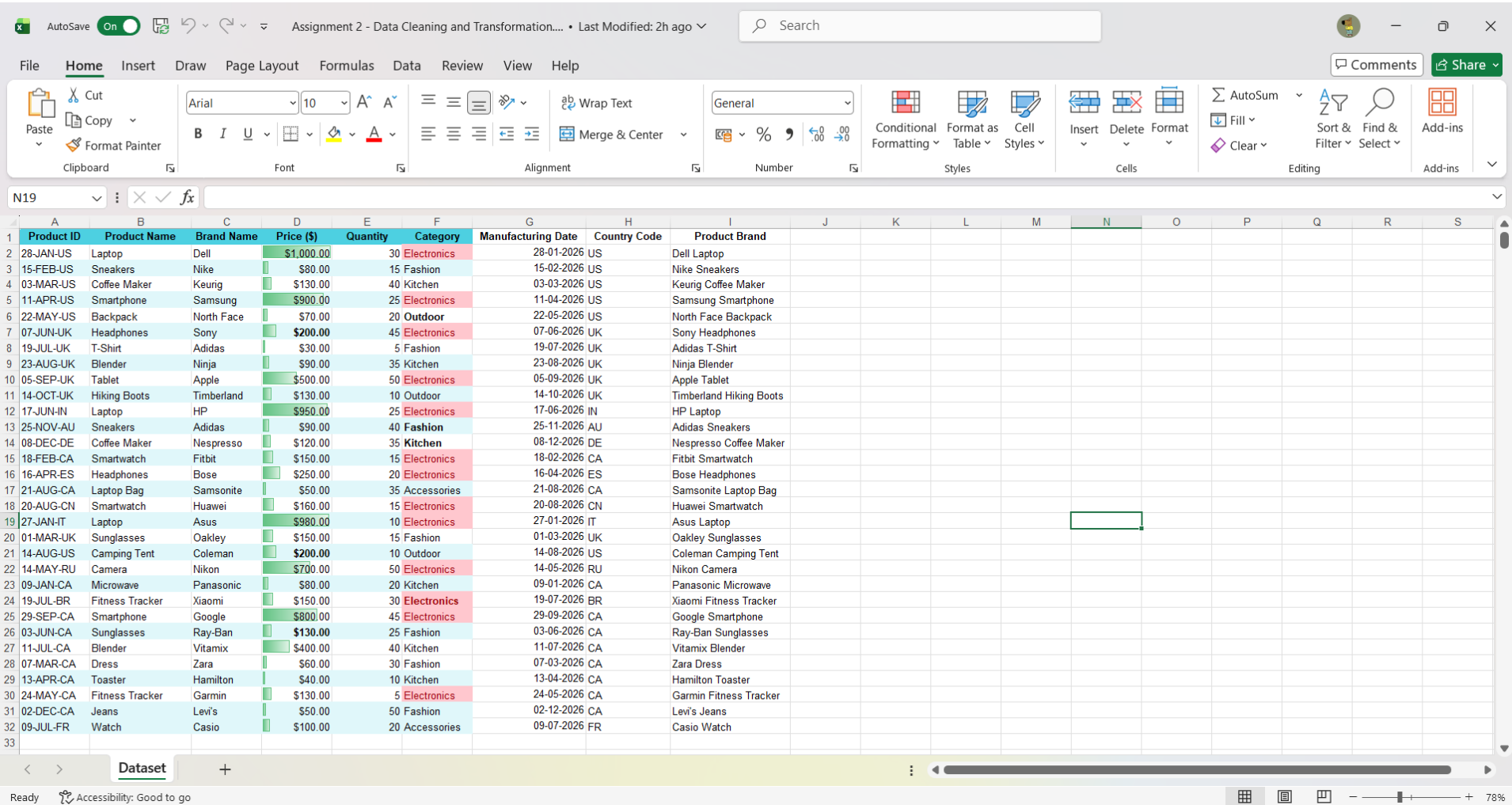
Figure 9: Number Formatting



Observation:

The **Price** column was formatted using the **Currency** format to display monetary values consistently. The **Manufacturing Date** column was formatted to the **DD-MM-YYYY** format, ensuring a standardized and easily readable date representation throughout the dataset.

Figure 10: Conditional Formatting Results



Observation:

Conditional formatting was applied to improve data visualization. **Data Bars** were added to the **Price** column to represent price values visually, making it easier to compare products. A custom conditional formatting rule was also applied to the **Category** column to highlight all products belonging to the **"Electronics"** category, allowing them to be identified quickly.